

CANONICAL LESSON

# Start With the Decision

Decision-centred foundations for accountable sector intelligence

|                 |                                   |
|-----------------|-----------------------------------|
| Production Unit | PU-B07-C01                        |
| Notebook        | V7_0_N01                          |
| Status          | Controlled educational draft v0.1 |
| Data boundary   | Synthetic instructional data only |

**Central decision:** Define the public decision, accountable owner, horizon, lawful intervention, and abstention rule before selecting a model.

## Learning outcomes

- By the end, you can: (1) define a public decision before choosing a model; (2) name the accountable owner and lawful intervention; (3) state the decision horizon and error trade-offs; (4) define abstention and escalation; and (5) produce a machine-readable decision contract.

## Why this matters

A technically accurate signal can still be unusable when nobody has authority, time, capacity, or a lawful intervention.

The analytical workflow is useful only when evidence, uncertainty, authority, and action are connected explicitly. The model supports professional judgement; it does not replace accountable institutional authority.

### 1. Why the decision comes first

- A forecast is useful only when a named institution can act within the forecast horizon.
- We therefore separate: signal, analytical interpretation, recommendation, authorization, and action.
- The model owns none of the last three.

### 2. A worked agricultural contract

- The analytical system may identify areas requiring field verification.
- It does not declare a food-security emergency or allocate resources automatically.

### 3. Loss is not the same as model error

- A false negative can be more consequential than a false positive, or vice versa.
- The relative cost is a policy judgement to document—not a number for the model to invent.

### 4. Actionability gate

- A high accuracy score is insufficient.
- A candidate passes only if the owner, intervention, horizon, authority, and operational capacity are all present.

### 5. Abstention is a valid output

- When required evidence is missing, the safest analytical output may be: insufficient evidence—refer for review.
- This protects decision-makers from false precision.

### 6. Cross-sector exercise

- Complete the contract below for an attendance/dropout early-warning service.
- Do not write 'the AI system' as the accountable owner.

### Worked cross-sector example

An attendance early-warning service identifies cases for authorized review; it does not impose sanctions or determine eligibility.

**Boundary:** The example supports learning and review. It does not constitute an official statistic, eligibility determination, enforcement action, clinical judgement, or operational authorization.

### Practice and reflection

- 1.
- Why must the decision horizon be defined before model selection?
- 2.
- Who owns the final decision?
- 3.
- Give two reasons to abstain.
- 4.
- Why is accuracy alone insufficient?

## Answer key

- 1.
- It determines whether data latency and forecast lead time permit intervention.
- 2.
- The named lawful institutional authority, never the model.
- 3.
- Examples: inadequate data quality; uncertain probability band; failed linkage; distribution shift.
- 4.
- Accuracy does not encode consequences, equity, calibration, actionability, authority, or operational capacity.

## Operational readiness checklist

- Purpose and decision owner are named.
- Evidence definitions, provenance, and lawful access are documented.
- Uncertainty and abstention are visible.
- Human review, escalation, monitoring, incident response, and retirement are assigned.
- Equity, accessibility, security, and sector-specific approval are complete.